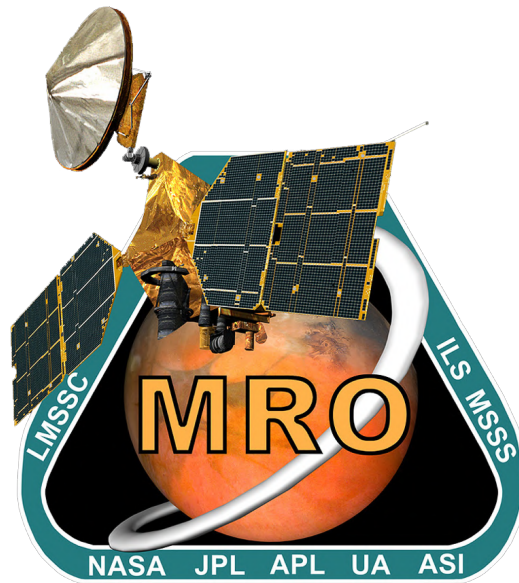




MRO Reverse Engineering Study

Assignment #3

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1 Tracking, Telemetry & Telecommand

The Mars Reconnaissance Orbiter (MRO) mission has been designed to capture an unprecedented amount of high-resolution data and transmit it in the form of compressed raw data from the spacecraft (S/C) to Earth. The full mission success criteria for data transfer was 26 terabits (Tb), the nominal Primary Science Phase (PSP) goal was 34 Tb, and over 50 Tb with the inclusion of Extended Mission (EM) and relay data. At the time of this report, MRO is in the EM and relay phase, having far surpassed both its expected mission lifetime and data transfer requirements (> 60 Tb).

Given the very high data transfer requirements and the Earth - S/C distance, an antenna with a very high gain is required for data transfer and communications. The S/C is also equipped with a pair of Low-Gain Antennas (LGAs) to enable low-data-rate communications to maintain basic Earth link during anomalies or critical phases. Additionally, the Electra Proximity Link Payload (Electra) system, operating in the Ultra High Frequency (UHF) band, is included to achieve the MRO mission goal of acting as a primary telecommunication node within the Mars network. Figure 1 shows a schematic of the entire telecommunications architecture implemented on the MRO, along with the placement of the antennas on the spacecraft body.

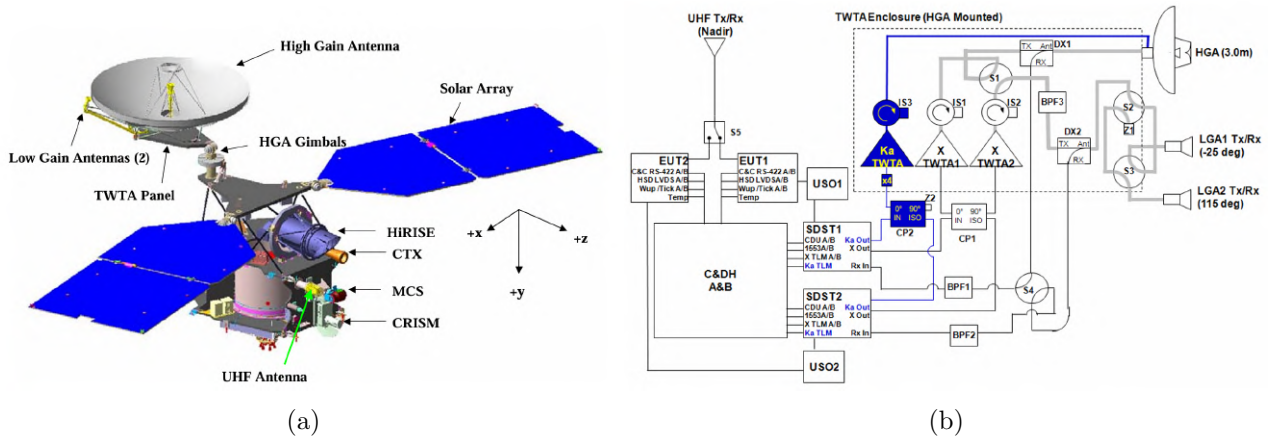


Figure 1: (a): Configuration of MRO. (b) Telecommunication scheme of MRO [9].

1.1 Architecture and rationale

1.1.1 High-Gain Antenna System

The High-Gain Antenna (HGA) serves as the primary mean of communication between Earth and the S/C, addressing the high gain requirement. It is equipped with three traveling-wave tube amplifiers (TWTAs): two in the X-band (as redundancy), and one in the Ka-band (see section 1.1.4). The high gain required is achieved by the use of reflectors or Direct Radiating Arrays (DRAs). While the latter provides the capability of electronic beam steering, it also introduces added complexity to the S/C, in addition to not being a technologically mature solution at the time of design. For similar reasons, along with considerations of reliability and signal integrity, a fixed parabolic reflector (shown in Figure 2a) was chosen over the alternative solutions such as deployable mesh antennas, which would also have been susceptible to the stressing aerobraking conditions. The HGA is highly directional, and must point at Earth for data transfer. As such, a two-axis gimbal mechanism was implemented to ensure continuous communication.

The HGA has a paraboloidal main reflector with axial displaced elliptical (ADE) geometry (3m in diameter) and an ellipsoidal subreflector (0.45m in diameter), shown in Figure 2b. The feed, placed in the focus of the system, has two components: a corrugated horn X-band feed and a disc-

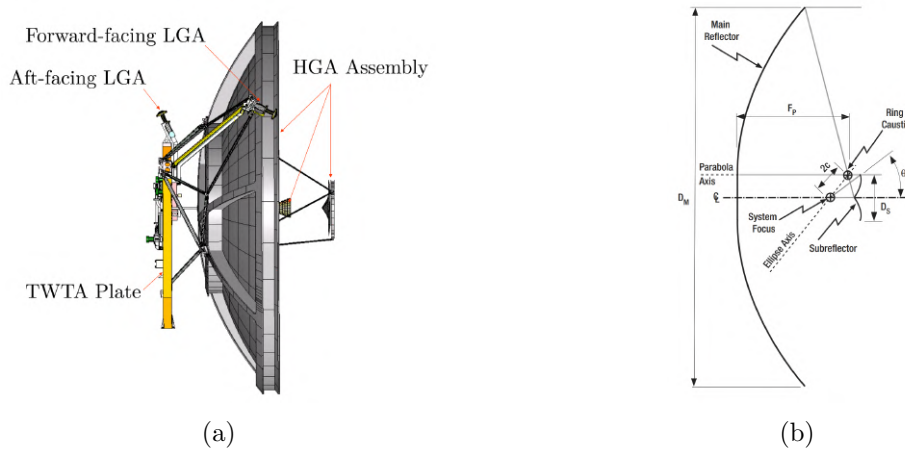


Figure 2: (a) The HGA, LGA, and TWTA[7]. (b) The ADE antenna geometry [5].

on-rod Ka-band feed [4]. The choice of corrugated horn for the X-band feed is critical to lower the cross-polarization, including those arising from the reflector configuration [1].

During aerobraking maneuvers, the HGA is locked in position in order to achieve the required drag surface, and telecommunications are happening through LGA1 [4].

1.1.2 Low Gain Antenna System

MRO is equipped with two Low-Gain Antennas (LGAs), designed for low-data-rate communication during critical mission phases such as launch, Mars Orbit Insertion (MOI), and safe mode operations. LGAs emit a broader radio beam, allowing for omnidirectional coverage at the expense of data rate. These horn-type antennas are essentially open waveguides with RF choke rings for pattern uniformity and side-lobe control. They are both mounted on the HGA, facing opposite directions. This configuration ensures continuous communication with Earth via the Deep Space Network (DSN), regardless of the spacecraft's orientation. The LGAs operate solely in the X-band frequency and do not support Ka-band capabilities. The forward-facing LGA1 is canted 25 degrees from the HGA boresight, whilst the aft-facing LGA2 is canted -115 degrees.

1.1.3 Electra

The Electra payload is a critical component for the relay operations of the S/C, and consists of a UHF transceiver and a low-gain UHF quadrifilar helix antenna. As for the NASA Mars Exploration Program (MEP), every future science orbiter will be required to host the standardized payload (P/L), in order to provide relay and navigation services to both present as future spacecraft part of MEP. MRO was the first orbiter to incorporate the Electra payload [6]. The transmitting and receiving frequencies depend on which mode the P/L is running, with Full Duplex transmitting in the 435-450 MHz band, whilst receiving in the 390-405 MHz band. If the P/L is in Half Duplex, the P/L both transmits and receives between 390 and 450 MHz.

In comparison to the science P/Ls of MRO, Electra is more specifically designed to avoid electromagnetic interference (EMI), which, if generated, would cause a degradation of the communications for the DSN. Specifically, at the default relay return link frequency of 401.59 MHz, the P/L is designed to withstand the effects of EMI. Electra includes a single nadir-pointing (vertically towards Mars) UHF LGA. The antenna shares the nadir deck with science payloads, some of which are sensitive to UHF frequencies. This proximity can lead to electromagnetic coupling between the payloads and the antenna, distorting its gain pattern.

1.1.4 Bands

X-band: Most of the communication with the Deep Space Network (DSN) has been executed at the X-band, 8.439 GHz for downlink (DL) and 7.183 GHz for uplink (UL) transmission. The X-band was chosen due to its ability to reliably communicate at long distances, resist atmospheric interference, support high data transfer rates and its long-standing heritage in deep space communications [4].

Ka-band: MRO was equipped with a Ka-band (32 GHz) transmitter to evaluate its potential for high-rate data return thanks to its wider allocated bandwidth compared to X-Band. During the cruise phase, the Ka-band system showed promising performance; on October 31, 2005, up to 133 Gbits were received at record speeds of 6 Mbps [4]. Although the Ka-band is more susceptible to atmospheric losses due to its higher frequency, it is less affected by solar interference compared to the X-band, as confirmed by previous in-flight tests. Despite providing these advantages, a Ka-band exciter anomaly occurred on May 26, 2006, just before the start of orbital operations. The project chose not to risk activating the system, and the operational demonstration was canceled for both the primary and extended missions [4].

UHF: UHF bands (435-450 MHz Full Duplex mode, 390-405 MHz only receiving mode) were selected for short-range relay communications between MRO and orbiters, landers, and rovers due to its high performance in the thin Martian atmosphere. At the aforementioned frequencies, atmospheric losses on Mars are almost negligible, even during high atmospheric activity periods. Additionally, the use of UHF ensures full compatibility with the Proximity-1 protocol, NASA's standard for relay operations at Mars [2].

1.1.5 Ground Section

The DSN was selected as the main communication network for the MRO mission, since it was specifically designed to support interplanetary missions by adopting high-gain antennas and providing the required link power to compensate for the stringent free-space losses caused by the large Earth–Mars distance. The DSN is composed of three stations at widely separated longitudes to provide continuous view from deep space (Goldstone in the USA, Madrid in Spain, and Canberra in Australia) [4].

These stations employ different types of antennas. The 70 m diameter antennas are primarily used during critical phases like Mars Orbit Insertion or safe mode operations due to their higher gain, pointing precision and sensitivity (signal acquisition is possible even at extremely low received power levels from the LGAs), as quantitatively described in the technical report [10].

the 34 m beam-waveguide (BWG) antenna subnet, and the 34 m high-efficiency (HEF) antenna subnet are used for routine communication sessions, such as cruise, science and relay phases due to the greater flexibility, lower operational cost, and easier scheduling provided by said subnets [4][8].

Other communication networks were used during the Launch and Early Orbit Phase (LEOP) phase, such as the Uchinoura Space Center (Japan), which received the first signal after separation [4].

1.1.6 Encoding and Modulation

The modulation of a carrier signal is performed inside the Small Deep Space Transponder (SDST), employing Phase-shift keying (PSK) instead of Frequency-shift keying (FSK), as PSK represents bits by shifting the phase of a single carrier frequency, whereas FSK uses multiple frequencies to represent bits (wider bandwidth), which is less suited for a deep space link. Binary phase-shift keying (BPSK) and Quadrature phase-shift keying (QPSK) are used at different phases and for different operational requirements [4]:

- **BPSK:** in a sub-carrier, or directly in the X-band signal carrier, the phase of the signal is shifted between 0° or 180°, to represent the bit state 0 or 1, BPSK is used in low-data-rate, emergency, or recovery phases.

- **QPSK**: four phase-shifting are used (0° , 90° , 180° , or 270°) to represent four possible bit pairs (00 01 10 11), QPSK is used for high-data-rate transmission during regular science operations[4].

The distance between Mars and Earth and the high data rate impose a very low signal-to-noise ratio (SNR), therefore powerful error-correction code (ECC) are needed. X-band and Ka-band communication systems employed the following coding schemes[4]:

- **Turbo Codes**: support bit rates above 32 kbps with block lengths of 8920 bits and code rates of $1/2$, $1/3$, $1/6$. They are favored in telemetry and science data links on X and Ka bands due to their near Shannon-limit performance, allowing high protection to high data rate for a given bandwidth. Turbo code encoding is used for robust error correction over long distances [9].
- **Reed-Solomon (RS)**: Up to 6.6 Mbps, used primarily for very high data rates in situations where MRO is close enough to Earth that coding gain is less important, due to an already strong signal, such as early cruise and Mars-Earth closest approach (low SNR);
- **Convolutional Concatenated**: Combination of $(7, 1/2)$ convolutional and RS codes, used in interplanetary mission where power is limited. This code is employed in emergency mode (34.38 bps) in short frames, and up to 3.3 Mbps in long frames during Ka-band downlink;

1.2 Con-Ops and Phases correlation

During the LEOP phase, only the LGAs were used to maintain basic communication, once the HGA was deployed, a stronger link with ground was established and the cruise phase began. During cruise, all communication systems were tested, but only LGA and HGA were used for routine data exchange, while most Ka-band experiments took place. In critical maneuvers such as Trajectory Correction Maneuver (TCM)s and MOI, only the LGAs were used due to HGA pointing constrains. During aerobraking, communication alternated between the two systems: the LGA near pericenter (highest criticality), and the HGA along the rest of the orbit. The Ka-band communications demonstration was planned to conduct activities during solar conjunction to compare simultaneous X and Ka-band telemetry downlinks. The DSN supported one 8 hour pass per day to a 34-m antenna during this period. Once in its operating orbit, MRO went through periodic occultations behind Mars, preventing Earth contact for about one third of orbit's time. Nevertheless, during the orbital mission, data was transmitted for 10–11 hours daily. In the PSP, MRO was allocated two 34-m passes per day (8 hours each) and three 70-m passes per week. The Electra UHF relay system became operational only in the dedicated relay phase [4].

System	Phases and ConOps	Band	Rationale
HGA	Cruise, Aerobraking, Science and Relay phases, Solar Conjunction (Ka-band tests)	X-band / Ka-band	High Precision High Datarate Dual-band (X/Ka) Capability
LGA	Safe mode, Launch Phase, TCMs , Aerobraking, MOI	X-band	Low Power Omnidirectional Coverage Emergency Link Support
Electra	Early Cruise Test, Relay Phase	UHF	Wide Beamwidth Mars Network Relay Node

Table 1: Relation with Operations, phases, modes

1.3 Reverse Sizing

This section aims at describing the reverse sizing procedure of the TMTC subsystems, focusing exclusively on the X-band communications with Earth during PSP, the phase with the highest data rate,

under the worst-case operation conditions. This section includes the input data and the assumptions for the analysis, followed by a presentation of the results for both uplink and downlink.

1.3.1 Downlink Sizing Data and Assumptions

The X-Band frequency used to send data from MRO to Earth is $f_{dl}=8439.55$ MHz according to guidelines provided by ITU, with a data rate $R_{dl} = 500$ kbps [4]. Taking into account the most critical condition, Earth-Mars distance was assumed at its highest (401.4 million km) during near-solar conjunction. The available power of the transmitting 3-m wide HGA is $P_{tx}=100$ W [4], with an electrical power to be provided to the antenna of 140 W, considering the amplifier efficiency recovered graphically ($\eta_{amp}=0.6$) and an aperture efficiency of $\eta_{HGA}=0.65$ due to its geometry. The system uses a BPSK modulation scheme ($\alpha_{mod}=1$) and a turbo code (8920, 1/2) encoding ($\alpha_{enc}=2$). While 1/6 rate turbo code would be more robust, the ground decoding only supports data rates up to 1.5 Mbps using turbo code [9]. The 10^{-5} BER leads to a minimum energy bit over noise power spectral density of 0.5 dB [12]. Some assumptions have also been taken to compute the losses affecting the link budget. The cable losses were estimated with a typical value of -1.0 dB (reported as minimal [4]) and the total pointing losses were taken from literature equal to -0.1 dB [11]. The atmospheric attenuation has been taken into consideration based on values available in the literature for an elevation angle of 10° and average clear skies ($CD = 0.25$) [3]. The Sun-Earth-Probe (SEP) angle was assumed sufficiently large ($> 1.5^\circ$) such that solar noise contribution is negligible. The system noise temperature was calculated using the procedure outlined in [8].

1.3.2 Uplink Sizing Data and Assumptions

The UL sizing procedure follows the same steps as the DL, with changes to some assumptions and data. The same distance is assumed as DL, as the same worst case cited above is considered for both UL and DL. The UL frequency is $f_{ul} = 7.183$ GHz, with a data rate of $R_{ul} = 4$ kbps, taking into account a margin around values found in the literature [4]. The ground station DSS 25 in Goldstone, which has an aperture of 34-m, an aperture efficiency of $\eta_{tx} = 0.65$, and a power capacity of 20 kW [11]. It was assumed that a power of $P_{tx} = 2$ kW is reserved for communicating with MRO. The modulation scheme for the UL is BPSK ($\alpha_{mod} = 1$), and 1/2 rate convolutional encoding ($\alpha_{enc} = 2$) is employed. For UL data transmission, a minimum bit error rate (BER) of 10^{-7} is set. Cable losses are assumed $L_c = -2$ dB. The atmospheric loss is estimated using $A_{zenith} = -0.04$ dB from [3] and an assumed elevation of 10° . The noise temperature was assumed as $T_{rx}=290$ K following the guidelines provided for the subsystem sizing.

1.3.3 Results Analysis

The results for both UL and DL are reported in Table 2. The overall link budget shows that the subsystem is well-sized for a reliable link both in UL and DL. As expected, free space losses are the driving losses as one of the main factors in MRO's link budget, due to the large distance S/C - Earth. The computed gains of the HGA antenna for the X-band for both UL and DL are similar to the ones reported in the literature [4].

In UL, high power is needed in order to compensate for the low gain of the receiving antenna on the S/C. The resulting E_b/N_0 and SNR are both well above the required minimums, granting solid reliability.

The DL case is much more critical, due to less available power. In addition, the data rate is considerably higher. These aspects make communications with Earth much more challenging. The computed values of E_b/N_0 and SNR are expectedly lower than the UL case. Both values are still above their respective minimums, but by a smaller margin than in the previous case, which makes the

DL sizing less reliable than that of UL, as any misapproximation could lead to overstepping the link budget constraints. The DL sizing would need more detailed modeling and data in order to obtain a more reliable result. In order to enhance the results computed in the DL case, one could reduce the data rate, increase the available power, or increasing gain of either antenna.

	UL	DL	Unit
Transmitted power (P_{tx})	2000	100	W
TX antenna gain (G_{tx})	66.28	46.60	dB
RX antenna gain (G_{rx})	45.20	67.68	dB
Cable losses (L_c)	-2	-1	dB
Pointing losses (L_{point})	-0.21	-0.1	dB
Atmospheric losses (L_{atm})	-0.22	-0.22	dB
Free space losses (L_{FS})	-281.02	-282.42	dB
Bandwidth (B)	40.49	61.46	dB
Net datarate (R_{net})	4	500	kbps
Gross datarate (R_{gross})	8	1000	kbps
RX system noise temperature (T)	290	33	K
Minimum Eb/N0 (with 3 dB margin)	8	3.5	dB
Minimum SNR (with 3 dB margin)	6.54	2.04	dB
Eb/N0	26.16	6.31	dB
SNR	24.70	4.85	dB

Table 2: MRO link budget

1.4 Mass and Power Budget

The TMTC subsystem mass and power budget are summarized in the table below considering the different components involved. The reported values highlight the importance of this subsystem into the design process of the whole satellite. The choice of gimbals for the HGA is justified by the need of continuous communications, even if it strongly affects the total mass budget. The power required by TWTAs is then necessary to cover the long distance with respect to Earth.

Assembly	Mass [kg]	Power [W]
X-band transponder	6.4	16
TWTAs	12.1	253
X-band & Ka-band antennas	22.6	-
HGA gimbals	45.0	14
Waveguides	8.3	-
Ultra-Stable Oscillator	1.7	5
UHF subsystem	11.5	71
Telecom total	107.7	359

Table 3: TTMTTC Subsystem Mass and Power Breakdown [4]

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Acronyms

ADE	axial displaced elliptical.	1, 2
BER	bit error rate.	5
BPSK	Binary phase-shift keying.	3, 5
DL	downlink.	3, 5, 6
DRA	Direct Radiating Array.	1
DSN	Deep Space Network.	2–4
ECC	error-correction code.	4
Electra	Electra Proximity Link Payload.	1, 2, 4
EM	Extended Mission.	1
EMI	electromagnetic interference.	2
FSK	Frequency-shift keying.	3
HGA	High-Gain Antenna.	1, 2, 4–6
ITU	International Telecommunication Union.	5
LEOP	Launch and Early Orbit Phase.	3, 4
LGA	Low-Gain Antenna.	1–4
MEP	Mars Exploration Program.	2
MOI	Mars Orbit Insertion.	2, 4
MRO	Mars Reconnaissance Orbiter.	1–6
P/L	payload.	2
PSK	Phase-shift keying.	3
PSP	Primary Science Phase.	1, 4
QPSK	Quadrature phase-shift keying.	3, 4
S/C	spacecraft.	1, 5
SDST	Small Deep Space Transponder.	3
SEP	Sun-Earth-Probe.	5
SNR	signal-to-noise ratio.	4, 5

TCM Trajectory Correction Maneuver. 4

TMTC Telemetry and Telecommand. 4, 6

TWTA traveling-wave tube amplifier. 1, 2, 6

UHF Ultra High Frequency. 1, 2

UL uplink. 3, 5, 6